Polychromatic control of electron orbitals in single ions

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Abstract

In this paper we analyse control of electron motion in single ions by a polychromatic laser field. We develop an analytic representation of wave functions for the electron motion. We analyse the problem of stability of motion in such fields. We relate this problem to Stokes data of certain multidimensional factorization problems with irregular singularities. We discuss the role of rational resonances in frequencies of the polychromatic field. In general, we show the role that Diophantine properties of the frequencies play in the stability analysis and in the spectrum of quasienergies, and provide the general framework for analysis of boundaries of stability regions in the space of field amplitudes. Based on this mathematical formalism, we extend the theory of Velocity Selective Coherent Population Trapping to the case of polychromatic fields. We show that there exist dark motional states for generic fields. We discuss the possibility to use these dark quasienergy states as temporal qubits of a universal quantum computer, realized on electron states of the single ion. We show that if the space of adiabatic transitions of the laser fields is large enough, there is an open set of fields for which spontaneous emission can be suppressed to any prescribed multipole order in a transition between dark quasienergy stability regions.

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1 Introduction.

There has been tremendous activity in quantum control and quantum computing on trapped ions. However, the gates in these systems are slow, on the order of microseconds. An alternative would be to use the electrons of a single atom for quantum computing, e.g. the higher electronic orbits in heavy atoms such as Ytterbium. On the first glance it seems impossible because the dipole emission would lead to quick decoherence in the electron state and thus would ruin the computation. We explore the possibility to preclude these transitions by applying a strong polychromatic laser field to the atom. We develop mathematical formalism for this problem. Based on the developed mathematical methods, we demonstrate the existence of dark quasienergy states and adiabatic transitions with spontaneous emission suppressed to arbitrary multipole order.

Important motivation for our study is the previous work on dark states in polychromatic setting [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]. In this work, the concept of dark states in polychromatic field was explored experimentally and theoretically. We extend this research as follows. First, we add the description of quasienergy states. We give mathematical evidence for the richness of these states in polychromatic field. We demonstrate, using Gap Labelling Theorem, that there are infinitely many quasienergy states. We then add evidence that many of these quasienergy states are in fact dark.

Previous work on Floquet engineering [13] motivated part of this work. Our addition to this discussion is more advanced formalism for the treatment of the eigenrvalue problem. We use generalization of [14, 15, 16, 17, 18, 19, 20, 21, 22, 23] in order to analytically describe stability regions. We note that it is recent advances in mathematical methods that make our work possible. We are dealing with generalized irregular singularities, in which the exponents are in general irrational numbers. In this numerically more challenging situation, Diophantine properties of laser frequencies play important role. We elucidate connections to the Gap Labelling Theorem and to the size of stability regions.

Important motivation for our study is the literature on the theory of Velocity Selective Coherent Population Trapping [24, 9]. However we note that the theory developed there (including [25] and more recent developments) is not sufficient for our purposes because it focuses on the regime where there is fluorescence, and thus the formalism that they use relies on density matrix.

The results of this paper are the following. In sections 23 we lay out general properties of ions in polychromatic field. We introduce the notion of quasienergy. We relate the equations to the previously considered case of irregular singularities without monodromy. We discuss the relation of quasienergy and the conditions on the Stokes matrices of the irregular singularity.

In section 4 we give a practical algorithm for calculation of the Stokes data. We explain the role of Diophantine approximations for the frequencies of the polychromatic field.

In section 5 we address stability of motion. The stability is related to Lyapunov exponents which we interpret as eigenvalues of the Stokes data. The boundaries of the stability region in the parameter space is given by the vanishing of the real part of these eigenvalues. We discuss how to lift the condition of rationality of frequencies and the role of Diophantine properties of frequencies. This leads us to the notion of Arnold tongues in the space of field amplitudes.

In section 6 we adapt Gap Labelling Theorem to our situation. This theorem was a great achievement in mathematics of Schrodinger operators [26, 27, 28]. We construct C* algebras appearing in the formulation of Gap Labelling Theorem and relate them to certain series built from solutions to quasienergy equations and KAM denominators.

In section 7 we give general representation of wave functions in almost periodic fields. Our representation explicitly shows the role of rational resonances between wave vectors. We use this representation in the following section for the analysis of motional states and of ionization in adiabatic transitions.

In section 8 we discuss motional states. These motional states are crucial for discussion of dark states. We generalize VSCPT theory and show the way how exact wave functions can in principle be constructed numerically. Our qualitative conclusion of the presence of Cantor set of Brilloun zones for the motional state. This is the computational resource that we use to realize qubits.

In section 9 we demonstrate that dark quasienergy states can be used as qubits of a quantum computer. Our analysis is based on identification of forbidden zones in the translation spectrum, and provides a generalization of the theory [24, 25].

In section 10 we develop the theory of engineering gates between the qubits introduced in 9. We show that under certain conditions adiabatic transitions between dark quasienergy states can be realized that suppress spontaneous emission to high multipole order.

2 Ions in polychromatic fields.

We consider the idealized problem of hydrogen-like ions in the polychromatic field

$$A_\mu(t,x) = \sum_{a,\pm} A_{a,\pm} exp(\pm i\omega_a t \pm ik_a x) \tag{1}$$

$$i\partial/\partial t\,\psi(t,x) = H_0(r)\psi(t,x) + A(t,x)\psi(t,x) \tag{2}$$

This equation can be either non relativistic approaximation, or the full Dirac equation. The reason for this is that consideration of strong fields is essential. We seek expansion

$$\psi(t,x) = \sum f_n(t)\psi_n(r) \tag{3}$$

where $H_0\psi_n(r) = E_n\psi_n(r)$. Then

$$i\frac{\partial f_n(t)}{\partial t} = E_n f_n + \sum A_{n,m,a,\pm} exp(\pm i\omega_a t) f_m(t) \tag{4}$$

We seek solutions as

$$f_n(t) = \sum exp(i(E + \sum N_a\omega_a)t) f_n(t) \tag{5}$$

where E is the quasienergy. In the case of single charge moving in the potential, the equations for disperion relations take the form

$$\psi(t,\vec{x}) = \sum \psi_n exp(i(\sum n_a\omega_a + \omega)t + i(\sum n_a\vec{k}_a + \vec{k})\vec{x}) \tag{6}$$

After the change of variables $\eta = exp(i\omega t), \xi_i = exp(ik_i x_i)$, where ω, k_i are auxiliary variables, the equations of motion takes the form

$$(\omega\gamma_0\eta\frac{\partial}{\partial\eta} + \sum k_i\gamma_i\xi_i\frac{\partial}{\partial\xi_i} + m)\psi(\eta,\xi_i) = ie\sum A_{a,\pm}\eta^{\alpha_a}\xi_i^{\beta_i}\psi(\eta,\xi_i) \tag{7}$$

This equation is of generalized irregular hypergeometric type [29, 18]. In contrast to the case considered in this literature, $\alpha_a, \beta_{a,i}$ are potentially non rational numbers. As we explain in the following, generalizations of the resurgence phenomena [16] take place in these equations.

In the case when there are several charges, the equations of motion take the form

$$(i\frac{\partial}{\partial t} + i\sum\gamma_i\frac{\partial}{\partial\vec{x}_i})\psi(t,\vec{x}_1,...,\vec{x}_n) = \sum ie_i A_{a,\pm} exp(\pm i\omega_a t \pm i\sum\vec{k}_a\vec{x}_i)\psi(t,\vec{x}_1,...,\vec{x}_n) + \gamma_0\sum U(\vec{x}_i - \vec{x}_j)\psi(t,\vec{x}_1,...,\vec{x}_n) \tag{8}$$

We made instantaneous approximation so that there is single time variable. In ultraintence field regimes, there are corrections that correspond to expansion in velocity. We consider two regimes: one in which there is confinement in the relative coordinate(i.e. non ionization regime) and one in which there is ionization so that $\psi(t,\vec{x}_1,\vec{x}_2) = \psi_1(t,\vec{x}_1)\otimes\psi_2(t,\vec{x}_2)$.

3 Single frequency case. Stokes phenomenon and quasienergy spectra.

In case there is single frequency, the equation takes the form

$$i\frac{\partial f(t)}{\partial t} = Ef + (A_+ exp(i\omega t) + A_exp(-i\omega t))f \tag{9}$$

where $f(t)$ is the vector function in the Hilbert space of the atom. After the change of variables $x = exp(i\omega t)$ we get

$$-\omega x\frac{f(x)}{dx} = Ef + (A_+ x + A_/x)f \tag{10}$$

The case when dimension of f is 2 was studied in [30]. The equation with one irregular singularity

$$\frac{df}{dx} = (A + \frac{B}{x})f \tag{11}$$

was studied in [19].

We consider the equation 10 in the complex x-plane. There are two singularities at $x = 0, \infty$. When x encircles any of these singularities, the function f gets a factor $f \to Sf$. S is the Stokes matrix. We perform an analytic study of S in the next section. The stability regions correspond to the vanishing of real parts of logarithms of eigenvalues of this matrix. This condition defines certain codimension one varieties in the space of parameters. In the 2-dimensional case, an asymptotic procedure was developed in [30] based on resurgence theory. In the simplified case with one irregular singularity, it was shown in [19] that Gelfand-Tsetlin coefficients play a role in explicit formulas for S. Our goal is to generalize these formulas.

The points $x = 0, \infty$ are both irregular points and as a result they have associated Stokes data. Due to the fact that there are no other singular points, the Stokes data for each of these points are equal. Quasienergies are given by eigenvalues of the monodromy matrix. The coefficients of the Schrodinger equation are periodic in time. The equation is translation invariant in time with this period. However, the solutions are not invariant. The solutions get a monodromy factor under these translations

$$f(t+T) = Sf(t) \ . \tag{12}$$

In x-space we get

$$S = Texp(\int_{\zeta} \Omega(x)dx) \tag{13}$$

where $\Omega(x) = E + (A_+ x + A_-/x)$. The contour ζ encircles the essential singularity at $x = 0$. From this representation, we see that the monodromy matrix is an analytic function of amplitudes $A_\pm$ and that the regions of stability $Re(\lambda_i) = 0$ are certain analytic curves. These curves give us boundaries of Arnold tongues in this case.

4 Multifrequency case. Rational resonances and KAM theory.

The general strategy in multi frequency case consists in considering rational approximations $\omega_a \approx P_a/Q$, see [31], and taking the limit $Q \to \infty$. Changing variables $x = exp(it/Q)$

$$-x/Qdf/dx = E_0 f + \sum A_{a,\pm} x^{\pm P_a} f \tag{14}$$

This is again an equation with two irregular singularities. We seek a solution in the form

$$f(x) = \sum f_n x^{E+n} \tag{15}$$

where E is the quasienergy. Then

$$-(E+n)/Qa_n = E_0 a_n + \sum A_{a,\pm} x_{n \pm P_a} \tag{16}$$

This is equation of Jacobi type [32, 33]. In block form

$$(n+E+B)/QF_n = AF_n + CF_{n-1} + DF_{n+1} \tag{17}$$

where $B = [-P_{a_0}, ..., P_{a_0}]$ and A is a matrix of size $(2P_{a_0})$.

$$(n+E+A)F_n = BF_{n-1} + CF_{n+1} \tag{18}$$

This is the matrix version of a Fibonacci system considered in [34]. In the latter paper, multifractal behavior of the spectrum was observed.

Proposition 4.1. *The monodromy matrix is an entire function of matrices $A_{a,\pm}$*

The proof of this fact consists in examining the series for the T-exponent

$$M = 1/n! \int_C \Omega(x_1)dx_1 \int^{x_1} \Omega(x_2)dx_2 ... \int^{x_{n-1}} \Omega(x_n)dx_n \tag{19}$$

After substitution of Ω we get

$$M = \sum_{a_1,\pm,...,a_n,\pm} A_{a_1,\pm} ... A_{a_n,\pm} M_{a_1,\pm,...,a_n,\pm} \tag{20}$$

where

$$M_{a_1,\pm,...,a_n,\pm} = 1/n! \int_C x_1^{\pm P_1/Q} dx_1 \int^{x_1} dx_2 x_2^{\pm P_2/Q} ... \int^{x_{n-1}} dx_n x_n^{\pm P_n/Q} \tag{21}$$

The only terms that contribute satisfy

$$\pm P_{i_1} \pm P_{i_2} ... \pm P_{i_m} + m = kQ \tag{22}$$

for some integer k. We get an exponential upper bound by the tree counting argument. This result can be compared with the analysis [30], where more detailed information was obtained in the simplest case of a 2-dimensional system.

5 Arnold tongues.

In this section we study the problem of stability of motion in our system 14. Our analysis will be based on Lyapunov exponents. The 2-dimensional case of systems of the above type with real dynamics was studied in [35, 36]. In these references, Lyapunov exponents were related to the rotation number. This simplified analysis substantially. The notion of rotation number is not defined in higher dimensions and non real dynamics. We therefore have to use the more general notion of Lyapunov exponents.

In the previous section, we defined the monodromy matrix in the case when all frequencies are rational multiples of a single fundamental frequency. In the general case, we define Lyapunov exponents as

$$\lambda_i = lim_{Q\to\infty}\frac{1}{Q}log(eig(M_Q)) \tag{23}$$

where M_Q is the rational approximation to the system, discussed in the previous section. It can be shown that the same quantity can be obtained as the set of Lyapunov exponents for the linear evolution system

$$\lambda_i = lim_{T\to\infty}\frac{1}{T}\log(Texp(\int^T \Omega(t)dt)) \ . \tag{24}$$

We define Arnold tongues as manifolds in the parameter space $A_{a,\pm}$ that correspond to $Re(\lambda_i) = 0$. Arnold tongues arise in the context of dynamics on the unit circle [37, 38] and Hamiltonian systems [39, 40]. The interpretation of these tongues is that they give the boundary of stability of the system.

Numerical computation of Arnold tongues by direct methods is computationally expensive. In the 2-dimensional case monodromy matrix is related to rotation angle [41, 42]. In this case, the position of the boundaries of Arnold tongues were computed [36, 43]. We remark on the definition of generalization of rotation number in higher dimensions. In the case of linear irregular systems, there is a parameterization exponential torus as discussed in [44]. We observe that in 2d case, the dynamics on this torus is related to the rotation number in the sense of Johnson [41]. The general notion of rotation number is tightly related to the dynamics on the exponential torus.

The case of systems with spatial dynamics is substantially more complicated. If we identify spatial degrees of freedom as dimensions, we formally have an infinite dimensional system. Rotation number and Lyapunov exponent must be defined in the Hilbert space of space dependent wave functions. Practically this means that the matrices in the equations of type 17 are infinite dimensional. They are computed as matrix elements of $\exp(i\vec{k}_a\vec{r})$ on the states of fixed quasienergy. We are treating this case in the next section.

6 Gap Labelling Theorem.

In the context discussed in the previous section, Gap Labelling theorem was discussed in [45, 43]. In recent developments, Gap Labelling was discussed in the context of Schroedinger equation [46, 47]. In our case, Gap Labelling is useful because it enumerates the position of tips of the Arnold tongues.

Gap Labelling Theorem has interpretation using C^* algebras [48, 26, 27]. However, the above cited papers do not directly rely on C^* algebraic context. It is the purpose of this section to clarify the nature of C^* algebras associated with the polychromatic Schrodinger equation. We make the following connection. The equation

$$x\frac{df}{dx} = \sum A_{a,\pm}x^{\pm\alpha_a}f \tag{25}$$

has a formal solution

$$f = \sum f_n x^{n+E} \tag{26}$$

where E is the quasienergy.

Identification 1: We identify the space of representations of the $C*$ algebra with the space f_n. The stable states (in the sense of KAM) can be interpreted as corners of the Choquet simplex of states of the C^* algebra acting on the vectors f_n. The eigenvalue equation

$$(\vec{n}\vec{\alpha}+E)f_n = \sum f_{n\pm\delta_a}A_{a,\pm},\ \vec{n} = (n_a) \tag{27}$$

has solutions formally represented as

$$f_n(\alpha,A) = \sum F(n;(n_0,k_0),...,(n,k_d);A)\frac{1}{(n_0\alpha+E)^{k_0}...(n_d\alpha+E)^{k_d}} \tag{28}$$

where $F(n;(n_0,k_0),...,(n,k_d);A)$ satisfies an eigenvalue equation that is independent of α,E.

Identification 2 The C^* algebra associated to the Schroedinger equation in almost periodic field is generated by direct sum of $A_{a,\pm}$ twisted by the denominators $1/(n\alpha+E)^k$. Note that this C^* algebra is sensitive to the Diophantine properties of α. We hypothesise that the growth properties of this algebra lead to the quantization of quasienergies E.

7 Represenation of wave function in almost periodic field

In this section we consider a representation of the Dirac wave function in Coulomb field in almost periodic filed $\sum A_{a,\pm}exp(\pm i\omega_a t \pm i\vec{k}_a\vec{x})$. The Dirac equation

$$(i\gamma_\mu\partial_\mu + m)\psi = \sum A_{a,\pm}exp(\pm i\omega_a t \pm i\vec{k}_a\vec{x})\psi \tag{29}$$

after Fourier transform takes the form

$$(\hat{q}+m)\psi(q) + \alpha\int d^3l\frac{\psi(q_0,\vec{l})}{(\vec{q}-\vec{l})^2} = \sum A_{a,\pm}\psi(q\pm k_a)\ . \tag{30}$$

Without any perturbation, the Dirac-Coulomb problem takes the form

$$(\hat{q}+m)\psi(q) + \alpha\int d^3l\frac{\psi(q_0,\vec{l})}{(\vec{q}-\vec{l})^2} = 0\ . \tag{31}$$

The solutions to this problem are of the form

$$\psi(q) = \frac{P(\vec{q})}{(\vec{q}^2+\lambda^2)^\gamma} \tag{32}$$

where $\gamma^2 = (j+1/2)^2 - (Z\alpha)^2$, and $P(q)$ is some polynomial. We consider the addition of a laser field in two stages.

Stage 1: We consider the static field $A(\vec{x}) = \sum A_{a,\pm}exp(\pm i\vec{k}_a\vec{x})$. Then we seek the solution in the form

$$\psi(\vec{q}) = \sum_{n\in\mathbb{Z}^d}\frac{P_n(\vec{q})}{(\vec{q}+\sum_a n_a\vec{k}_a)^2+\lambda^2)^\gamma} \tag{33}$$

where the sum is over the lattice $\mathbb{Z}^d$. We see that there are such formal solutions because the set of quadrics $(\vec{q}+\sum_a n_a\vec{k}_a)^2 + \lambda^2 = 0$ is closed with respect to both vanishing cycle development in the system $(\vec{p}+\sum_a n_a\vec{k}_a)^2+\lambda^2 = 0, (\vec{p}-\vec{q})^2 = 0$ [49], and shifts by $\pm\vec{k}_a$.

Stage 2: When the field is almost periodic $A(t,\vec{x}) = \sum A_{a,\pm}exp(\pm i\omega_a t \pm i\vec{k}_a\vec{x})$, then we seek a solution of the form

$$\psi(t,\vec{x}) = \sum\psi_m(\vec{x})exp(i(\sum m_a\omega_a + E)t) \tag{34}$$

where E is the quasienergy and $\psi_m(\vec{x})$ are formal sums of the above type.

Stage 3: When we have adiabatic process $A(t,\vec{x}) = \sum A_{a,\pm}(t)exp(\pm i\vec{k}_a\vec{x})$, then we seek solutions in the form of formal sums

$$\psi(t,\vec{q}) = \sum_{n\in\mathbb{Z}^d}\frac{P_n(t,\vec{q})}{(\vec{q}+\sum_a n_a\vec{k}_a)^2+\lambda^2)^\gamma} \tag{35}$$

where $P_n(t,\vec{q})$ now has temporal dependence. From this representation, we see that formal solutions have continuous dependence on the external almost periodic field.

8 Dark states.

In the above discussion we assumed that the nucleus is infinitely heavy. In this section we analyse the case of the nucleus of finite mass. We will see that this case presents a new class of dark states. The situation considered here is a close analogy of [24, 25]. However, for our analysis it is impossible to make the approximations considered in these references. In particular, we do not assume that the system has only two states. Moreover, we do not assume smallness of the field. Only after we lift these approximations it is possible to draw conclusions about the spectrum of states with forbidden spontaneous emission. While treatment of charge motion in a single plane field is available, this formalism is impossible to generalize to polychromatic case. We start with states of free charge in a plane polychromatic field. We assume for simplicity that the charge is described by Dirac equation

$$(i\hat{\partial} + m + \sum \hat{A}_{a,\pm} exp(\pm ik_a x))\psi(x) = 0 \tag{36}$$

We seek solutions of the form

$$\psi(x) = \sum \psi_n exp(i(\sum n_a k_a + l)x) \tag{37}$$

where l is a wave vector. We get the equation

$$(-(\sum n_a k_a + l) + m)\psi_n = \sum \hat{A}_{a,\pm}\psi_{n\pm\delta_a} \tag{38}$$

This is an equation of block Jacobi type and much of our discussion in sec. 6 carries over to this case as well. The difference with the previous case is the presence of the parameter $\vec{l}$. We obtain the dispersion relation $l_0 = l_0(\vec{l})$ in this parameter. For rationally independent k_a, we on general grounds expect Cantor-like Brillouin zones. In addition to our analytic representation in sec. 6, the above Discrete Schrodinger Equation(DSE) of block Jacobi type was discussed [33, 50, 32, 51], with similar conclusions. Versions of Gap Labelling Theorem adapted to the case of Schrodinger operators on trees are discussed in [52]. From the comparison with this case, we see that after suitable renormalization, eigenfunctions are expressed in terms of multivariable orthogonal polynomials. The spectral function $\omega(\vec{k})$ is of devil staircase type [1]. In multiple dimensions, we observe the classical one dimensional devil staircase behavior if we take line section of the graph of the function $\omega(\vec{k})$.

The general phenomenon observed in 1d systems is the presence of a Cantor set in the space of parameters. This Cantor set is the set of parameters where the rotation number changes. Similar behavior is exhibited by the pseudo frequency $\omega(\vec{p})$: there is a measure zero set in $\vec{p}$ where $\omega(\vec{p})$ changes. This behavior of $\omega(\vec{p})$ has the following implication:

Definition 8.1. *(Dark state) Consider the emission of a photon with momentum $\vec{k}$. If there is no components C_1, C_2 of the complement of the Cantor set with $\omega(\vec{p}) \in C_1$ and $\omega(\vec{p} - \vec{k}) \in C_2$, then such emission could not happen. This corresponds to the fact that the emission occurs from the total system of atom + laser. Such states will be called dark states.*

In the above definition, the photon energy k must be the difference of two quasienergies of the electron system. Dark states arise when the following condition holds: for any two quasienergy states E_1, E_2, for any allowed direction of emission of a photon with energy $|\vec{k}| = E_1 - E_2$, there are no allowed motional zones in the spectrum such that $\vec{k}$ connects these two zones.

The previous discussion leads us to the following proposal. Cosider dark motional states with high quasienergy. These states have forbidden emission because of the above mechanism that excluded spontaneous emission. These states are highly nonstationary and have internal dynamics. Based on the Gap Labelling, and on general properties of spectra (presence of Cantor set of jumps in the spectrum $\omega(\vec{k})$) we expect an infinite set of dark motional states. These states are the resource of quantum computation. They can be used as atomic versions of qubits.

Note that the state space is explicitly infinite dimensional in the systems we consider. Infinite dimensionality is an important resource that makes many computations possible [53, 54, 55]. The question lies in the numerical size and experimental stability of these states. This question concerns the size of the Arnold tongues as we discussed in section 5. The computation involves the analysis of isomonodromy Stokes matrices as a function of field amplitudes $A_{a,\pm}$.

We note that already hydrogen-like ions possess most of the properties necessary for realization of quantum computation. The discussion of the properties of spectra and of dark states is valid in this simple case. The presence of an almost periodic field makes the system non integrable and sensitive to the bifurcation patterns observed in the dependence of Stokes data on field parameters. The presence of multiple electrons makes the system much richer. In particular, there are quantum analogies of KAM tori [56, 57, 58]. These quantum KAM tori were only partially studied experimentally and mostly in the model situations [59, 60, 61, 62]. For physical systems, such as helium, only very preliminary results are available [63]. We hope that

[1] Reader can see pictures of such sets in [43]

development of methods of precision control such as [64] will lead to precision studies of quantum KAM tori for helium-like ions, which would open the way for utilization of the inherently chaotic states as a quantum computing resource.

We have to compare the above discussion of dark motional states with the theory of Velocity Selective Coherent Population Trapping(VSCPT). This theory has a long history [24, 25]. While this theory gives qualitative explanation of phenomena, for practical purposes of the use of dark states for computation, we need a quantitative theory. For example, the theory listed in the above publications is not sufficient for quantitative analysis of forbidden motional zones. The full analysis, as we discussed above, involves theory of spectra of block Jacobi matrices with the linear ergodic part [33]. We demonstrated how to analyse this spectrum in simple cases and related it to the simultaneous rational approximation of frequencies of the laser beams. We mentioned the basic ingredients of such quantitative theory above.

9 Temporal qubits. Gates.

In this section we further investigate the use of dark quasienergy states as qubits. Since these states are inherently dynamical, we call the corresponding qubits temporal qubits. In the previous sections we considered quasienergy states in the presence of polychromatic fields with fixed amplitudes and frequencies. The space of qubits is numbered by according to the Gap Labelling Theorem by the rational quasienergy numbers

$$|\vec{n}>=|p_a/q_a>=\psi_{p_a/q_a}(t,\vec{x}) \tag{39}$$

where p_a/q_a are the rational numbers appearing in Gap Labelling Theorem. It will be advantageous to select broad Arnold tongues, as these tongues correspond to stability regions in the space of field amplitudes. Gates connecting quasienergy states will be realized as adiabatic trajectories

$$A_{a,\pm} \to A_{a,\pm}(t)\ . \tag{40}$$

The trajectory that corresponds to the transition 40

$$|\vec{n},A>\to|\vec{n'},A'>\ . \tag{41}$$

These trajectories must be carefully engineered in order to reduce the loss of information in such transitions. While in the literature mostly studied are effects of decoherence, in our case there is another important mechanism for information loss, namely, possible spontaneous emission in the course of the adiabatic transition. We will show how to minimize this potential source of information loss.

The quasienergy numbers $\vec{n'}$ can in general be different than $\vec{n}$. The key condition that such transition must satisfy is the absence of spontaneous emission in the adiabatic process. The space of fields $A_{a,\pm}(t)$ is essentially constrained by the laser system. Our goal is to provide general considerations for optimal choice of the trajectory $A_{a,\pm}(t)$. This problem has been extensively studied in model problems [65].

The conditions for $A_{a,\pm}(t)$ are two fold:

1) The motional bands 8 change in such a way that the spontaneous decay into lower quasienergy spectrum remain forbidden during the transition.

2) The multipole $\vec{m}$ matrix elements have vanishing time average

$$\frac{1}{T}\int_0^T <\vec{n'}(t)|\vec{m}|\vec{n}(t)>\to 0 \tag{42}$$

Our goal is to develop ab initio formalism that is suitable for precise description of multielectron atoms in string almost periodic field. One can compare this ambitious goal with existing formalisms [66, 67, 68, 69, 70, 71]. We use the modern formalism of resurgence theory [72, 73, 74, 75, 76, 77, 78] and related developments in complex analysis [79, 80, 81]. For our ab initio study, it is convenient to use Dirac multiparticle equation, because the solution class is in fact not much more complicated than the one for non relativistic approximation.

$$(i\hat{D}(A(x_i))-m_i)\psi(x_1,...,x_n)=0 \tag{43}$$

In this system of equations $D_\mu(A(x_i))$ is the covariant derivative at point x_i. The field $A(x_i)$ is the full field

$$A(x_i)=\sum A_{a,\pm}exp(\pm ik_a x_i)+A^C(x_i) \tag{44}$$

Here $A^C(x_i)$ is the field created by atomic system at the point x_i. This field is the self consistent solution of the following equation

$$\Box A^C_\mu(x_i) = J_\mu(x_i) \tag{45}$$

where the self consistent current is

$$J_\mu(x_i) = i\sum \int_{y_1,...,y_{i-1},y_{i+1},...,y_n} \bar{\psi}(y_1,...,x_i,...,y_n) e_i \gamma^{(i)}_\mu \psi(y_1,...,x_i,...,y_n) \tag{46}$$

Theorem 9.1. *The system 43 of coupled equations has a resurgent solution with double exponential asymptotics at spatial infinity in each of the variables $\vec{x}_i$. Quantization is the condition that for each coordinate the real axis $x_{i,s}, s = 1,2,3$ is the Stokes line. Quasienergy values are the values at which the Stokes matrix at these Stokes lines have vanishing real part.*

Proof: First, we can choose equal time wave functions $\psi(t,\vec{x}_1,...,\vec{x}_n))$ because the operators in 43 are first order evolution operators. We use regularized Hylleraas representation at finite distance

$$\psi(x_1,...,x_n) = \sum \psi_{l_{i,j}}(t) \prod \frac{1}{|\vec{x}_i - \vec{x}_j|^{\alpha_{i,j}+l_{i,j}}} \tag{47}$$

Such representation exists because the powers $|\vec{x}_i - \vec{x}_j|^{\beta_{i,j}}$ are expanded depending on $|\vec{x}_i|/|\vec{x}_j|$ and matching conditions can be used on the diagonals to relate expansion coefficients in these regions. At spatial infinity we have expansion

$$\psi(x_1,...,x_n) = \sum \psi(n_{a,s}) exp(i(\sum n_{a,s} k_a + l)x_s) \tag{48}$$

where l is the quasimomentum. Similar to our analysis in section 8 we obtain energy bands in the motional spectrum $l_0(\vec{l})$.

The quantization condition amounts to the requirement that the solution is an entire function of coordinates, including the quadrics $(\vec{x}_i - \vec{x}_j)^2 = 0$, that is no more than double exponential at infinity. The matching of the expansions 47 and 48 lead to the equations for the Stokes matrices at $(\vec{x}_i - \vec{x}_j)^2 = 0$, $\vec{x}_i \to \infty$ in the complex domain. From these representations we conclude the existence of solutions in the specified form.

□

Rk: There is a toy model were analysis can be carried further

$$i\frac{\partial}{\partial t}\phi(t,x,y) = -\Delta\phi + \frac{1}{p(x,y)}\phi + \sum A_{a,\pm}(t) exp(\pm ik_{a,1}x + \pm ik_{a,2}y)\phi \tag{49}$$

where $p(x,y)$ is a generic polynomial. The theorem says that in this case ϕ is an entire function of x,y with double Stokes structure at infinity, which is obtained by matching conditions at $p(x,y) = 0, x \to \infty$. In this example, it means that $\phi(t,x,y) = p(x,y)e(t,x,y)$, where $e(t,x,y)$ is an entire function. Note the connection to the theory of nodal surfaces [71]. The nodal surfaces are time dependent in this case.

We now turn to the design of gates realized by the polychromatic system. Gates on trapped ions were considered in many papers [82, 65]. Gate corresponds to time varying laser field

$$A = \sum A_{a,\pm}(t) exp(\pm ik_a x) \tag{50}$$

where $A_{a,\pm}$ is a slowly varying function with different asymptotics at $\pm\infty$. One example is furnished by

$$A_{a,\pm}(t) = A_{a,\pm,-\infty} + \sum \Delta A_{a,\pm,i} erf((t-t_i)/\sigma_i) \tag{51}$$

where $erf(x)$ is the error function.

Theoretical study of higher dark states was performed in [83, 9, 11]. We extend this study to non stationary strong fields. Motional state of the ion plays a key role in this discussion. In [83] the following state was used

$$\psi = |g^+, p-k> - |g^-, p+k> \tag{52}$$

We generalize this wave function to complete temporal state $\psi(t,\vec{x}_1,...,\vec{x}_n)$. In polychromatic fields, one may argue [11], that functions are of the form

$$\psi = \sum |g, p + \sum n_a k_a> \ . \tag{53}$$

However, when k_a are rationally independent, the points $\sum n_a k_a$ fill the space densely. This is a well known phenomenon in resurgence theory [84]. The explicit numerical computation therefore must rely on the full analytic machinery of resurgence theory. In this paper, we only prove the existence of desired behavior, i.e. the existence of non emitting states. The states connect two quasienergy states for different values of field parameters. In addition, in the course of such transition, spontaneous emission can be suppressed to any desired multipole order, as long as the laser field has enough control degrees of freedom:

Theorem 9.2. *(existence of non stationary dark states) If the space of laser field parameters 50 has sufficiently large dimension, there exists a path between the initial $A_{a,\pm,-\infty}$ and final $A_{a,\pm,+\infty}$ field configurations such that along this path the condition 42 holds for a given multipole order N.*

Proof:

The key to our argument is continuous dependence of the time dependent wave function on the parameters of the field. This can be observed as follows. The Fourier transform of the exact relativistic equation in a time dependent field transforms the equations of motion into integral equations with a kernel that equals the Fourier transform of the field. If this kernel has holomorphic dependence on the field parameters, the same is true about the solutions. This can be seen explicitly when the field is of exponential resurgent type.

□

Rk: In practice, solution of the equations 42 require the computation of PDE analogies of Voros invariants [85]. Significant extension of currently available methods is necessary for this task. We laid out the beginnings of this procedue in sec. 6 in 0-dimensional case and pointed out the relevance of the C^* algebra of terms built on small denominators. In the multidimensional case, there is analogous C^* algebra, but this time twisted by the coordinate dependent part. This coordinate depenedent part is of no more than doubly exponential limit at spatial infinity.

Rk: In the above discussion, the non recoild condition played a key role. The argument that we put forth is that of existence of a manifold of field configurations such that in the transition the multipole emission from the system $ion + laser$ is essentially forbidden for energy reasons: if a photon is emitted, then, due to recoil, the wave function of the ion in the laser field changes in such a way that the energy of the total system $ion + laser$ grows, which makes such emission impossible.

10 The probability of non ionization

Another very important aspect of stability analysis is the suppression of ionization. Ionization in strong periodic field has been considered in publications including [86]. It has only numeric solution. In the case of hydrogen an analysis can be found in [87]. The amplitude of ionization in the adiabatic transition $A(1) \to A(2)$ is given by

$$A(T) = \int_0^T dt < \psi_{m,E}(\vec{k}) \otimes \psi_{M,E}(\vec{-k}) | \psi(t) > \tag{54}$$

Here, $\psi(t)$ is the wave function of motional eigenstate considered above. $\psi_{m,E}, \psi_{M,E}$ are wave functions of quasifree electron and nucleus propagating separately. The condition that the probability of such transition is vanishing is $A(T)/T \to 0$.

The treatment that we are going to provide should be contrasted with the familiar treatment of High Harmonic Generation [88, 89]. The treatment in these papers is essentially classical. However, we are interested in fine details of the motion. In some cases, such treatment is already available in the literature [90, 91]. It can be observed that in some of these simulations there are zeros in the ionization current as function of the amplitude. This corroborates our conlcusion.

We formulate a model problem in which ionization can be studied. It is known [92, 93] that there is a marked difference between 1,2 and higher dimensional cases. It is necessary to consider the the problem in 3 dimensions. It is also necessary to consider non confining potentials. Coulomb potential is the simplest such potential. For example, while quadratic oscillator still exhibits Cantor spectra in polychromatic fileds, the non ionization is trivial in this case. The problem of the choice of basic functions in the presence of polychtromatic fields $exp(i\vec{k}_a\vec{x})$ is known to lead to double exponential behavior at spatial infinity [94]. It is also known [95] that arbitrary small additions to Coulomb potential can lead to dense point spectrum.

The the minimal set of functions sufficient for the choice of basis in this situation comprises resurgent functions of multiple variables with uniform double exponential asymptotics at spatial infinity and at most regular singualrities at the diagonals in the configuration space. Elements of the theory of such functions can be found in [96, 97, 98]. On general grounds that there is analogy of Stokes structure at spatial infinity for these doubly exponential functions. The Stokes matrices are infinite dimensional in this case, with indices running over quasienergy states, but the conclusion of their analytc dependence on field

amplitudes is still valid. This argument shows that there is non empty region in the space of amplitudes that corresponds to non ionization in the adiabatic transition

Theorem 10.1. *(existence of non ionizing transitions) If the parameter space of laser fields of the form 50 is large enough, there is manifold of field configurations such that in the transition $A_{a,\pm,1} \to A_{a,\pm,2}$ the ionization probability is zero.*

Rk: from practical perspective, it is important to find the region in the parameter space such that it is stable to noise in the laser field. The size of this region is a generalization of the stability analysis that we carried out in sec 6. Full numerical analysis requires computation of the Stokes matrices for the spatially dependent problem.

11 Conclusion

In this paper we studied qualitatively the problem of control of electron orbitals by polychromatic laser fields. We devloped key mathematical ingredients for exact treatement of such problems. We connected the problem to the known case of ordianry differential equations with irregular singularities and showed how the stability of electron motion is related to the Stokes data in this model problem. We discussed the case arbitray incommensurate frequencies and showed how Diophantine approximations to these frequencise control approximation scheme for quasienergy computations. We showed that the problem in this case reduces to a block version of Jacobi problem. We argued on general grounds that there is a Cantor set in the spectrum of the problem.

We devloped a formalism for treatement of electron motion in polychromatic fields beyond rotating wave approximation. In fact, our general formalism is applicable in fields of arbitrary strength. The conclusion of the presence of Cantor sets in the spectrum remains valid in this case as well.

We then proceeded to the analysis of motional states in polychromatic field. We developed analytic representation of wave functions in such motional states, extending previously developed formalism for VSCPT. Based on this representation, we demonstrated the presence of dark motional states. Together with the realization of Cantor sets in the spectrum of quasienergy, these states provide a large computational resource for quantum computing. We explained how to realize qubits in the Hilbert space of the problem. Then we discussed the problem of realization of gates between such states. We argued theoretically that it is possible to realize transtions between the dark states such that along such transitions the spontaneous radiation can be suppressed to high order. We hope that in this way we demonstrated theoretical possibility of universal computing on single multielectron ions.

12 Declaration statements

The author declares no conflict of interest. No data was consudmed/created in this research.

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References

[1] H Metcalf. Dark states and de broglie wave optics. *Acta Physica Polonica A*, 93(1):147–157, 1998.

[2] M Doery, M Widmer, J Bellanca, E Vredenbregt, T Bergeman, and H Metcalf. Energy bands and bloch states in 1d laser cooling and their effects on the velocity distribution. *Physical review letters*, 72(16):2546, 1994.

[3] L Liu, MJ Bellanca, and H Metcalf. Evolution of coherent dark states. *Physical Review A*, 65(4):043401, 2002.

[4] M Widmer, MR Doery, MJ Bellanca, WF Buell, TH Bergeman, and HJ Metcalf. High-velocity dark states in velocity-selective coherent population trapping. *Physical Review A*, 53(2):946, 1996.

[5] MR Doery, MT Widmer, MJ Bellanca, WF Buell, TH Bergeman, H Metcalf, and EJD Vredenbregt. Population accumulation in dark states and subrecoil laser cooling. *Physical Review A*, 52(3):2295, 1995.

[6] MR Doery, R Gupta, T Bergeman, H Metcalf, and EJD Vredenbregt. Bichromatic velocity-selective coherent population trapping. *Physical Review A*, 51(3):2334, 1995.

[7] Harold Metcalf. Colloquium: Strong optical forces on atoms in multifrequency light. *Reviews of Modern Physics*, 89(4):041001, 2017.

[8] Harold Metcalf. Laser cooling as a form of optical pumping in the quantum domain of atomic motion. *Physica Scripta*, 1997(T70):57, 1997.

[9] J Hack, L Liu, M Olshanii, and H Metcalf. Velocity-selective coherent population trapping of two-level atoms. *Physical Review A*, 62(1):013405, 2000.

[10] X Miao, E Wertz, MG Cohen, and H Metcalf. Strong optical forces from adiabatic rapid passage. *Physical Review A—Atomic, Molecular, and Optical Physics*, 75(1):011402, 2007.

[11] Leonid Yatsenko and Harold Metcalf. Dressed-atom description of the bichromatic force. *Physical Review A—Atomic, Molecular, and Optical Physics*, 70(6):063402, 2004.

[12] VM Entin, EA Yakshina, DB Tretyakov, II Beterov, and II Ryabtsev. Spectroscopy of the three-photon laser excitation of cold rubidium rydberg atoms in a magneto-optical trap. *Journal of Experimental and Theoretical Physics*, 116(5):721–731, 2013.

[13] Martin Holthaus. Floquet engineering with quasienergy bands of periodically driven optical lattices. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 49(1):013001, 2015.

[14] Xiaomeng Xu. Regularized limits of stokes matrices, isomonodromy deformation and crystal basis. *arXiv preprint arXiv:1912.07196*, 2019.

[15] Naihuan Jing, Ming Liu, and Alexander Molev. Isomorphism between the r-matrix and drinfeld presentations of quantum affine algebra: type c. *Journal of Mathematical Physics*, 61(3), 2020.

[16] Anton Alekseev, Andrew Neitzke, Xiaomeng Xu, and Yan Zhou. Wkb asymptotics of stokes matrices, spectral curves and rhombus inequalities. *Communications in Mathematical Physics*, 405(11):269, 2024.

[17] Philip P Boalch. G-bundles, isomonodromy, and quantum weyl groups. *International Mathematics Research Notices*, 2002(22):1129–1166, 2002.

[18] Jinghong Lin and Xiaomeng Xu. Explicit evaluation of the stokes matrices for certain quantum confluent hypergeometric equations. *Journal of Geometry and Physics*, 209:105364, 2025.

[19] Xiaomeng Xu. Irregular riemann–hilbert correspondence, alekseev–meinrenken dynamical r-matrices, and drinfeld twists. *International Mathematics Research Notices*, 2018(23):7420–7456, 2018.

[20] Xiaomeng Xu. Stokes phenomenon and yang–baxter equations. *Communications in Mathematical Physics*, 377(1):149–159, 2020.

[21] Xiaomeng Xu. Stokes phenomenon and reflection equations. *Communications in Mathematical Physics*, 398(1):353–373, 2023.

[22] Xiaomeng Xu. Stokes phenomenon, gelfand–zeitlin systems and relative ginzburg–weinstein linearization. *Advances in Mathematics*, 338:237–265, 2018.

[23] Philip Boalch. Irregular connections and kac-moody root systems. *arXiv preprint arXiv:0806.1050*, 2008.

[24] Alain Aspect, Ennio Arimondo, R Kaiser, N Vansteenkiste, and C Cohen-Tannoudji. Laser cooling below the one-photon recoil energy by velocity-selective coherent population trapping: theoretical analysis. *Journal of the Optical Society of America B*, 6(11):2112–2124, 1989.

[25] François Bardou. *Lévy statistics and laser cooling: how rare events bring atoms to rest*. Cambridge University Press, 2002.

[26] Jean Bellissard. Gap labelling theorems for schrödinger operators. In *From number theory to physics*, pages 538–630. Springer, 1992.

[27] Jerome Kaminker and Ian Putnam. A proof of the gap labeling conjecture. *Michigan Mathematical Journal*, 51(3), 2003.

[28] Wencai Liu and Xiaoping Yuan. Spectral gaps of almost mathieu operators in the exponential regime. *Journal of Fractal Geometry*, 2(1):1–51, 2015.

[29] Werner Balser. Multisummability of formal power series solutions of partial differential equations with constant coefficients. *Journal of Differential Equations*, 201(1):63–74, 2004.

[30] Gerald V Dunne and Mithat Ünsal. Wkb and resurgence in the mathieu equation. In *Resurgence, physics and numbers*, pages 249–298. Springer, 2017.

[31] Kostya Khanin, João Lopes Dias, and Jens Marklof. Multidimensional continued fractions, dynamical renormalization and kam theory. *Communications in mathematical physics*, 270(1):197–231, 2007.

[32] Jacob S Christiansen, Barry Simon, and Maxim Zinchenko. Finite gap jacobi matrices, i. the isospectral torus. *Constructive Approximation*, 32(1):1–65, 2010.

[33] Injo Hur and Christian Remling. Ergodic jacobi matrices and conformal maps. *Mathematical Physics, Analysis and Geometry*, 15(2):121–162, 2012.

[34] Jake Fillman, Yuki Takahashi, and William Yessen. Mixed spectral regimes for square fibonacci hamiltonians. *Journal of Fractal Geometry*, 3(4):377–405, 2016.

[35] Henk Broer and Mark Levi. Geometrical aspects of stability theory for hill's equations. *Archive for rational mechanics and analysis*, 131(3):225–240, 1995.

[36] Henk Broer and Carles Simó. Hill's equation with quasi-periodic forcing: resonance tongues, instability pockets and global phenomena. *Boletim da Sociedade Brasileira de Matemática-Bulletin/Brazilian Mathematical Society*, 29(2):253–293, 1998.

[37] Araceli Bonifant, Xavier Buff, and John Milnor. Antipode preserving cubic maps: the fjord theorem. *Proceedings of the London Mathematical Society*, 116(3):670–728, 2018.

[38] Kuntal Banerjee, Xavier Buff, Jordi Canela, and Adam Epstein. Tips of tongues in the double standard family. *Nonlinearity*, 34(12):8174, 2021.

[39] Rafael de la Llave and Yannick Sire. An a posteriori kam theorem for whiskered tori in hamiltonian partial differential equations with applications to some ill-posed equations. *Archive for Rational Mechanics and Analysis*, 231(2):971–1044, 2019.

[40] Raphaël Krikorian, Jing Wang, Jiangong You, and Qi Zhou. Linearization of quasiperiodically forced circle flows beyond brjuno condition. *Communications in Mathematical Physics*, 358(1):81–100, 2018.

[41] Russell A Johnson. Exponential dichotomy, rotation number, and linear differential operators with bounded coefficients. *Journal of differential equations*, 61(1):54–78, 1986.

[42] David Damanik, Meirong Zhang, and Zhe Zhou. The rotation number for almost periodic potentials with jump discontinuities and δ-interactions. In *Annales Henri Poincaré*, volume 25, pages 1359–1397. Springer, 2024.

[43] Joaquim Puig and Carles Simó. Analytic families of reducible linear quasi-periodic differential equations. *Ergodic Theory and Dynamical Systems*, 26(2):481–524, 2006.

[44] Martin Klimeš. Stokes phenomenon and confluence in non-autonomous hamiltonian systems. *Qualitative theory of dynamical systems*, 17(3):665–708, 2018.

[45] Joaquim Puig and Carles Simó. Resonance tongues and spectral gaps in quasi-periodic schrödinger operators with one or more frequencies. a numerical exploration. *Journal of Dynamics and Differential Equations*, 23(3):649–669, 2011.

[46] David Damanik, Íris Emilsdóttir, and Jake Fillman. The schwartzman group of an affine transformation. *Journal of Spectral Theory*, 13(4):1281–1296, 2023.

[47] Long Li, David Damanik, and Qi Zhou. Cantor spectrum for cmv matrices with almost periodic verblunsky coefficients. *Journal of Functional Analysis*, 283(12):109709, 2022.

[48] J Bellissard, R Benedetti, and JM Gambaudo. Spaces of tilings, finite telescopic approximations and gap-labelling, to appear in commun. *Math. Phys*, 2004.

[49] Paul Seidel. More about vanishing cycles and mutation. *Symplectic geometry and mirror symmetry (Seoul, 2000)*, pages 429–465, 2001.

[50] Injo Hur, Matt McBride, and Christian Remling. The marchenko representation of reflectionless jacobi and schrödinger operators. *Transactions of the American Mathematical Society*, 368(2):1251–1270, 2016.

[51] Jacob S Christiansen, Barry Simon, and Maxim Zinchenko. Finite gap jacobi matrices, iii. beyond the szegő class. *Constructive Approximation*, 35(2):259–272, 2012.

[52] Alexander Aptekarev, Sergey Denisov, and Maxim Yattselev. Self-adjoint jacobi matrices on trees and multiple orthogonal polynomials. *Transactions of the American Mathematical Society*, 373(2):875–917, 2020.

[53] William Slofstra. Tsirelson's problem and an embedding theorem for groups arising from non-local games. *Journal of the American Mathematical Society*, 33(1):1–56, 2020.

[54] Miguel Navascués, Stefano Pironio, and Antonio Acín. A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New Journal of Physics*, 10(7):073013, 2008.

[55] Marius Junge, Miguel Navascues, Carlos Palazuelos, David Perez-Garcia, Volkher B Scholz, and Reinhard F Werner. Connes' embedding problem and tsirelson's problem. *Journal of Mathematical Physics*, 52(1), 2011.

[56] Vassili Georgievich Gelfreich and Vladimir Fedorovich Lazutkin. Splitting of separatrices: perturbation theory and exponential smallness. *Russian Mathematical Surveys*, 56(3):499, 2001.

[57] Sergej Kuksin and Jürgen Pöschel. Invariant cantor manifolds of quasi-periodic oscillations for a nonlinear schrödinger equation. *Annals of Mathematics*, pages 149–179, 1996.

[58] Yuan Wu and Xiaoping Yuan. On the existence of full dimensional kam torus for fractional nonlinear schrödinger equation. *Journal of Applied Analysis & Computation*, 10(2):771–794, 2020.

[59] S Yoshida, CO Reinhold, P Kristöfel, and J Burgdörfer. Exponential and nonexponential localization of the one-dimensional periodically kicked rydberg atom. *Physical Review A*, 62(2):023408, 2000.

[60] Simon A Gardiner, JI Cirac, and Peter Zoller. Quantum chaos in an ion trap: the delta-kicked harmonic oscillator. *Physical review letters*, 79(24):4790, 1997.

[61] Korana Burke, Kevin A Mitchell, Shuzhen Ye, and F Barry Dunning. Chaotic ionization of a stationary electron state via a phase space turnstile. *Physical Review A—Atomic, Molecular, and Optical Physics*, 88(1):013408, 2013.

[62] Gabriel G Carlo, Giuliano Benenti, and Dima L Shepelyansky. Dissipative quantum chaos: transition from wave packet collapse to explosion. *Physical review letters*, 95(16):164101, 2005.

[63] Alfred S Schlachter. The search for quantum chaos: From celestial mechanics to the helium atom. *Radiation Physics and Chemistry*, 75(12):2159–2164, 2006.

[64] Alexey Grinin, Arthur Matveev, Dylan C Yost, Lothar Maisenbacher, Vitaly Wirthl, Randolf Pohl, Theodor W Hänsch, and Thomas Udem. Two-photon frequency comb spectroscopy of atomic hydrogen. *Science*, 370(6520):1061–1066, 2020.

[65] Yotam Shapira, Ravid Shaniv, Tom Manovitz, Nitzan Akerman, Lee Peleg, Lior Gazit, Roee Ozeri, and Ady Stern. Theory of robust multiqubit nonadiabatic gates for trapped ions. *Physical Review A*, 101(3):032330, 2020.

[66] MS Safronova, WR Johnson, and UI Safronova. Relativistic many-body calculations of the energies of n= 2 states for the berylliumlike isoelectronic sequence. *Physical Review A*, 53(6):4036, 1996.

[67] Charlotte Froese Fischer, Michel Godefroid, Tomas Brage, Per Jönsson, and Gediminas Gaigalas. Advanced multiconfiguration methods for complex atoms: I. energies and wave functions. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 49(18):182004, 2016.

[68] Wojciech Cencek, Konrad Patkowski, and Krzysztof Szalewicz. Full-configuration-interaction calculation of three-body nonadditive contribution to helium interaction potential. *The Journal of chemical physics*, 131(6), 2009.

[69] Igor M Savukov, Dmytro Filin, Pinghan Chu, and Michael W Malone. Relativistic configuration-interaction and perturbation theory calculations for heavy atoms. *Atoms*, 9(4):104, 2021.

[70] Sebastian Höfener, Nils Schieschke, Wim Klopper, and Andreas Köhn. The extended explicitly-correlated second-order approximate coupled-cluster singles and doubles ansatz suitable for response theory. *The Journal of Chemical Physics*, 150(18), 2019.

[71] Tony C Scott, Arne Lüchow, Dario Bressanini, and John D Morgan III. Nodal surfaces of helium atom eigenfunctions. *Physical Review A—Atomic, Molecular, and Optical Physics*, 75(6):060101, 2007.

[72] Claude Mitschi, David Sauzin, and David Sauzin. Resurgent functions and alien calculus. *Divergent Series, Summability and Resurgence I: Monodromy and Resurgence*, pages 173–271, 2016.

[73] Jean Écalle. Six lectures on transseries, analysable functions and the constructive proof of dulac's conjecture. In *Bifurcations and periodic orbits of vector fields*, pages 75–184. Springer, 1993.

[74] Jean Ecalle and Bruno Vallet. The arborification-coarborification transform: analytic, combinatorial, and algebraic aspects. In *Annales de la Faculté des sciences de Toulouse: Mathématiques*, volume 13, pages 575–657, 2004.

[75] Jean Ecalle. Recent advances in the analysis of divergence and singularities. *Normal Forms, Bifurcations and Finiteness Problems in Differential Equations*, pages 87–186, 2003.

[76] Jean Ecalle. A tale of three structures: the arithmetics of multizetas, the analysis of singularities, the lie algebra ari. *Differential equations and the Stokes phenomenon*, 17:89–146, 2002.

[77] Claude Mitschi, David Sauzin, Michèle Loday-Richaud, and Éric Delabaere. *Divergent series, summability and resurgence*, volume 2153. Springer, 2016.

[78] David Sauzin. Mould expansions for the saddle-node and resurgence monomials. In *Renormalization and Galois theories. Selected papers of the CIRM workshop, Luminy, France*, pages 83–163, 2006.

[79] Walter Bergweiler, Alexandre Eremenko, and Lasse Rempe. Second order linear differential equations with a basis of solutions having only real zeros. *Journal d'Analyse Mathématique*, 152(1):53–108, 2024.

[80] Walter Bergweiler and Jie Ding. On the boundary of an immediate attracting basin of a hyperbolic entire function. *Journal of the London Mathematical Society*, 111(3):e70085, 2025.

[81] David Martí-Pete, Lasse Rempe, and James Waterman. Eremenko's conjecture, wandering lakes of wada, and maverick points. *Journal of the American Mathematical Society*, 2025.

[82] Anders Sørensen and Klaus Mølmer. Quantum computation with ions in thermal motion. *Physical review letters*, 82(9):1971, 1999.

[83] S Kulin, Y Castin, M Ol'shanii, E Peik, B Saubaméa, M Leduc, and C Cohen-Tannoudji. Exotic quantum dark states. *The European Physical Journal D-Atomic, Molecular, Optical and Plasma Physics*, 7(3):279–284, 1999.

[84] Eric Delabaere, Hervé Dillinger, and Frédéric Pham. Exact semiclassical expansions for one-dimensional quantum oscillators. *Journal of Mathematical Physics*, 38(12):6126–6184, 1997.

[85] André Voros. The return of the quartic oscillator. the complex wkb method. In *Annales de l'IHP Physique théorique*, volume 39, pages 211–338, 1983.

[86] Peter Lambropoulos, Paul Maragakis, and Jian Zhang. Two-electron atoms in strong fields. *Physics Reports*, 305(5):203–293, 1998.

[87] Mitsuko Murakami, Oleg Korobkin, and Marko Horbatsch. High-harmonic generation from hydrogen atoms driven by two-color mutually orthogonal laser fields. *Physical Review A*, 88(6):063419, 2013.

[88] Maciej Lewenstein, Ph Balcou, M Yu Ivanov, Anne L'huillier, and Paul B Corkum. Theory of high-harmonic generation by low-frequency laser fields. *Physical Review A*, 49(3):2117, 1994.

[89] AM Zheltikov. Keldysh parameter, photoionization adiabaticity, and the tunneling time. *Physical Review A*, 94(4):043412, 2016.

[90] Alexey Gorlach, Ofer Neufeld, Nicholas Rivera, Oren Cohen, and Ido Kaminer. The quantum-optical nature of high harmonic generation. *Nature communications*, 11(1):4598, 2020.

[91] Dmitry A Telnov and Shih-I Chu. Ab initio study of the orientation effects in multiphoton ionization and high-order harmonic generation from the ground and excited electronic states of h 2+. *Physical Review A—Atomic, Molecular, and Optical Physics*, 76(4):043412, 2007.

[92] Leonid Parnovski and Alexander V Sobolev. Bethe-sommerfeld conjecture for periodic operators with strong perturbations. *Inventiones mathematicae*, 181(3):467–540, 2010.

[93] Leonid Parnovski. Bethe–sommerfeld conjecture. In *Annales Henri Poincaré*, volume 9, pages 457–508. Springer, 2008.

[94] NB Delone and Vladimir P Krainov. Energy and angular electron spectra for the tunnel ionization of atoms by strong low-frequency radiation. *Journal of the Optical Society of America B*, 8(6):1207–1211, 1991.

[95] Sergei Nikolaevich Naboko. Dense point spectra of schrodinger and dirac operators. *Teoreticheskaya i Matematicheskaya Fizika*, 68(1):18–28, 1986.

[96] Walter Bergweiler and Alexandre Eremenko. Quasiconformal surgery and linear differential equations. *Journal d'Analyse Mathématique*, 137(2):751–812, 2019.

[97] YM Chiang and RG Halburd. On the meromorphic solutions of an equation of hayman. *Journal of mathematical analysis and applications*, 281(2):663–677, 2003.

[98] Mashael Alhamed, Lasse Rempe, and Dave Sixsmith. Geometrically finite transcendental entire functions. *Journal of the London Mathematical Society*, 106(2):485–527, 2022.